BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SUKKUR

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2017

CHEMISTRY

Past Paper 2017 • Paper Code: A

Retyped faithfully from the original board paper — same questions, no changes.

Section - C (Multiple Choice Questions)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** A

Note: Q.16: Choose the correct answer from the given options and write the same in your answer copy. Note: Cutting and Over writing is not allowed.

Q1. The sum of the atomic masses of all atoms in a formula unit of a substance is called _____

(A) Empirical formula
(B) Molecular formula
(C) Molecular mass
(D) Formula mass

Q2. The most abundant and useful halogen is _____

(A) Bromine
(B) Fluorine
(C) Iodine
(D) Chlorine

Q3. During experiment Rutherford tried to pass a beam of _____ particles through a very thin gold metal foil.

(A) Gama
(B) Alpha
(C) Beta
(D) X-rays

Q4. The boiling point of Sulphur is _____

(A) 454°C
(B) 464°C
(C) 434°C
(D) 444°C

Q5. The minimum energy required to remove electron from a gaseous atom in it ground state is called:

(A) Electron affinity
(B) Ionization energy
(C) Electro negativity
(D) Atomic radius

Q6. Iron metal obtained from blast furnace is impure and is called as _____

(A) Cast iron
(B) Pig iron
(C) Steel
(D) Wrought iron

Q7. The density of ice at 0°C is about _____

(A) 0.998 g/cm3
(B) 0.918 g/cm3
(C) 0.819 g/cm3
(D) 0.889 g/cm3

Q8. The process in which solid directly changes to gas is called _____

(A) Evaporation
(B) Melting
(C) Sublimation
(D) Boiling

Q9. Sodium hydrogen carbonate is used in medicine which of the effects it causes in the stomach _____

(A) Increase the basicity
(B) Increase the acidity
(C) Neutralize the acidity
(D) Decrease the basicity

Q10. In suspension, the suspended particles are generally of the size of:

(A) 1 nm
(B) 10 nm
(C) 100 nm
(D) 1200 nm

Q11. Fraction of petroleum containing C21 – C40 and boiling range is above 400°C is _____

(A) Light oil
(B) Heavy oil
(C) Paraffin wax
(D) Bitumen

Q12. Fused Silica which is also known as quartz glass is used for making:

(A) Concrete or cement
(B) Silicon carbide
(C) Optical lenses and prisms
(D) absorbent

Q13. The number of replaceable (OH) ions present in a molecule of base, is called _____

(A) Acidity
(B) Basicity
(C) Neutral
(D) Hydrolysis

Q14. Gaseous Oxygen is about _____ times denser than air.

(A) 1.4
(B) 1.3
(C) 1.2
(D) 1.1

Q15. 1 mole of methane is burnt in 2 moles of O2 then heat energy is released _____

(A) 890 KJ/mol
(B) 880 KJ/mol
(C) 870 KJ/mol
(D) 860 KJ/mol

Q16. The electrolyte dilute H2SO4 used in lead storage battery have concentration of _____

(A) 10%
(B) 20%
(C) 30%
(D) 40%

Q17. _____ described affectionately as the father of modern chemistry.

(A) Jabir Ibn Haiyan
(B) J- Black
(C) J - Priesly
(D) Robert Boyle

Section - A (Short Answer Questions)

Time Allowed: 2.40 Hours | **Maximum Marks:** 40

Note: NOTE:- ATTEMPT ANY EIGHT (08) QUESTIONS. EACH QUESTION CARRIES (05) MARKS.

Q1. How methane is prepared in Laboratory?

میتھین گیس کی لیبارٹری میں تیاری بیان کریں۔

Q2. What is water of crystallization?

قلمی پانی سے کیا مراد ہے؟

Q3. Define allotropy. Write the names of chief allotropic forms of Carbon?

ایلوٹروپی کی تعریف کریں۔ کاربن کے اہم ایلوٹروپک فارمز کے نام لکھیں۔

Q4. Florine, Chlorine, Bromine, Iodine and Astatine are placed in same group. Why? Also mention the name of the group.

فلورین، کلورین، برومین، آئیوڈین اور ایسٹیٹین ایک ہی گروپ میں رکھے گئے ہیں۔ کیوں؟ گروپ کا نام بھی لکھیں۔

Q5. What are three common stainless steel and their composition?

عام سٹینلیس اسٹیل کی تین اقسام کون سی ہیں اور ان کی ترکیب بیان کریں۔

Q6. What is Diffusion? Explain on the basis of kinetic molecular theory?

نفوذ کسے کہتے ہیں؟ حرکی مالیکیولی نظریے کی بنیاد پر وضاحت کریں۔

Q7. How is Hydrogen chloride is manufactured commercially by direct combination of H2 and Cl2 gases?

ہائیڈروجن کلورائیڈ، H2 اور Cl2 گیسوں کے براہ راست اتصال سے صنعتی طور پر کیسے تیار کی جاتی ہے؟

Q8. What is meant by the term saturated solution? Describe how you would make up a saturated solution of copper sulphate in water.

سیر شدہ محلول سے کیا مراد ہے؟ پانی میں کاپر سلفیٹ کا سیر شدہ محلول کیسے بنائیں گے، وضاحت کریں۔

Q9. You are given three closed gas jars, one contains Oxygen, another one contains carbon dioxide and the third one contains Hydrogen. Describe tests you would carry out to label each gas jar correctly.

آپ کو تین بند گیس جار دیے گئے ہیں، ایک میں آکسیجن، دوسرے میں کاربن ڈائی آکسائیڈ اور تیسرے میں ہائیڈروجن گیس ہے۔ ہر ایک جار کی درست لیبلنگ کے لیے آپ کس طرح جانچ کریں گے؟

Q10. Define pH? Explain why a neutral solution is said to have a pH of (7) seven?

pH کی تعریف کریں۔ غیر جانبدار (تعمیلی) محلول کی pH سات (7) کیوں ہوتی ہے؟

Q11. Write down the atomic number, mass number and numbers of proton, neutron and electron in:

(a) 18Ar40

(b) 25Mn55

Q12. Write note on the:

نوٹ لکھیں:

(a) Electron Affinity

الیکٹران ایفینیٹی

(b) Atomic Radii

ایٹمی نصف قطر

Section - B (Long Answer Questions)

Maximum Marks: 28

Note: NOTE: ATTEMPT ANY TWO (02) QUESTIONS. QUESTION NO: FIFTEEN (15) IS COMPULSORY.

Q13.

(a) State and explain Faraday's First Law of Electrolysis.

(b) What is hydrogen bonding? What types of forces either intra molecular or inter molecular forces are present in hydrogen bonding?

Q14.

(a) Discuss Rutherford's gold metal foil experiment. What did it tell about the structure of atom?

(b) Describe structure and uses of some of the common plastics.

Q15. Balance the following chemical equations and write the type of chemical equation.

مندرجہ ذیل کیمیائی مساواتوں کو متوازن کریں اور کیمیائی عمل کی قسم لکھیں۔

(i) $Ca + H2O \rightarrow Ca(OH)2 + H2$

(ii) $CH4 + O2 \rightarrow CO2 + H2O$

(iii) $Fe2O3 + CO \rightarrow Fe + CO2$

(iv) $KClO3 \rightarrow KCl + O2$

(v) $PbS + O3 \rightarrow PbSO4$

(vi) $NaOH + H2SO4 \rightarrow Na2SO4 + H2O$

(vii) [illegible in source scan] $\rightarrow CO2 + SO2 + H2O$